

Unlocking Behavioral Intelligence in Airline Loyalty Programs

Canadian airline loyalty cohort · 16,700 members · 2012–2018 · XGBoost churn model + K-Means segmentation

TOP BUSINESS FINDINGS

- 01

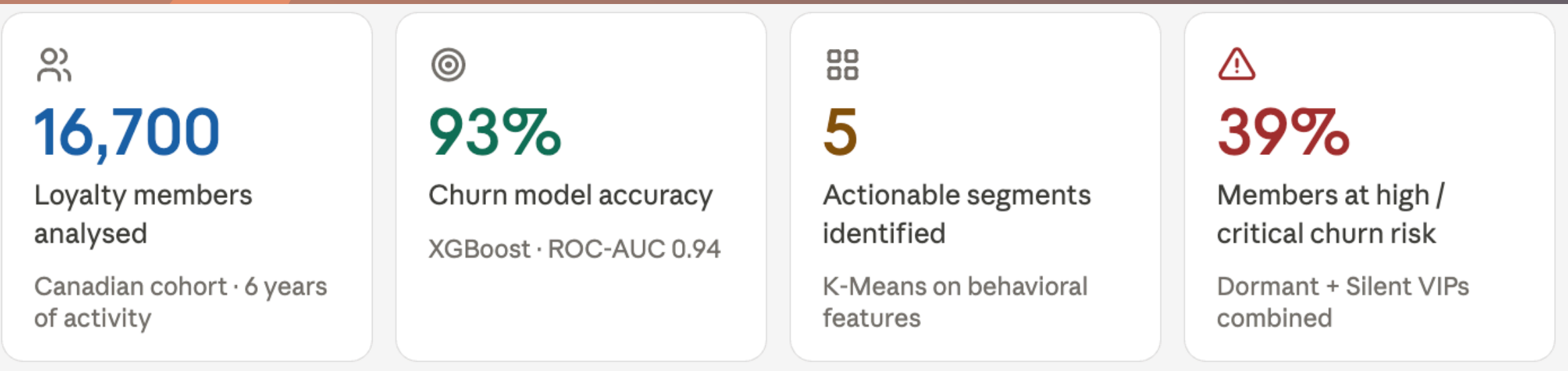
Churn model achieved **93% accuracy and ROC-AUC of 0.94** using behavioral features — flight frequency and redemption activity were the strongest predictors.
- 02

70% of churn risk originates from just 2 segments — Dormant Members (41% churn rate) and Reward Seekers (29% churn rate) — while representing ~45% of the base.
- 03

Silent VIPs hold \$27,539 avg CLV — 4.4× higher than any other segment — yet show declining activity. A single segment carries outsized revenue risk.
- 04

Segment-specific retention interventions — triggered at inactivity milestones, not quarterly calendars — can reduce expected churn by **15–20%** in the top two risk cohorts.

DATASET & MODEL AT A GLANCE



This report presents a data-driven behavioral intelligence framework built across three interlocked objectives:

- Churn Prediction** — Who is about to stop engaging, and can we know before they formally cancel?
- Customer Segmentation** — What does the membership base actually look like when you sort by behavior rather than just CLV?
- Smart Retention** — For each customer type, what specific intervention should fire, when, and why?

CHURN DEFINITION & FEATURE ENGINEERING

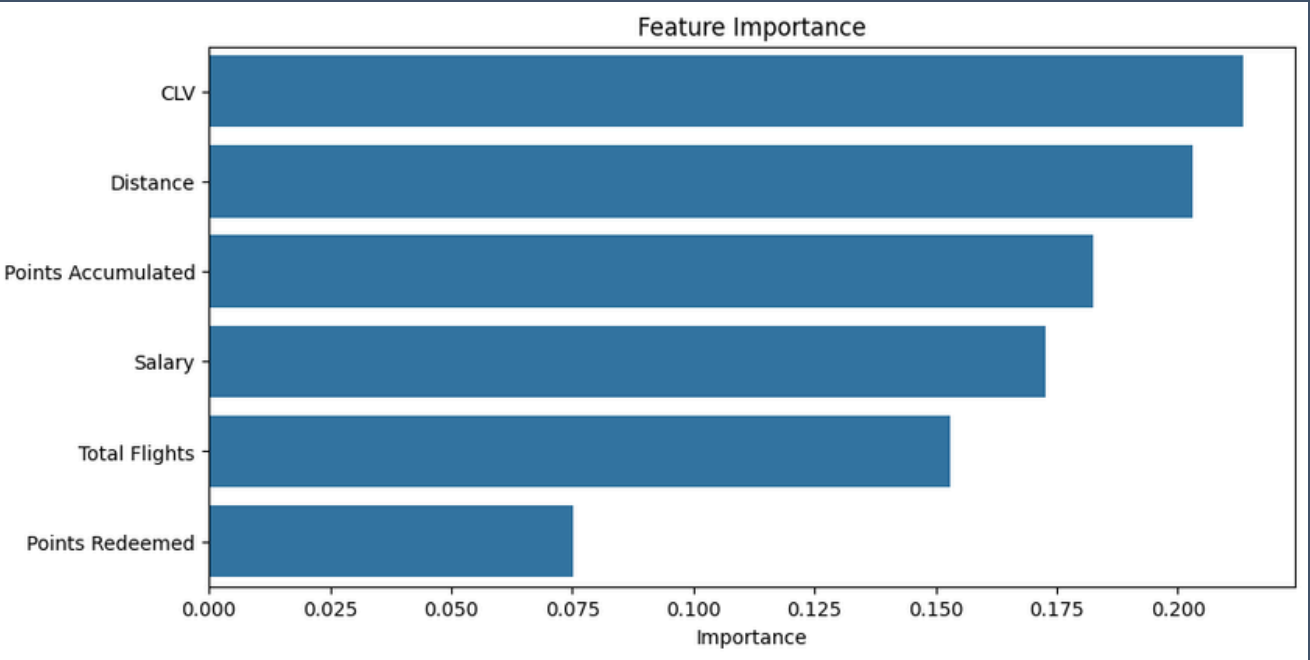
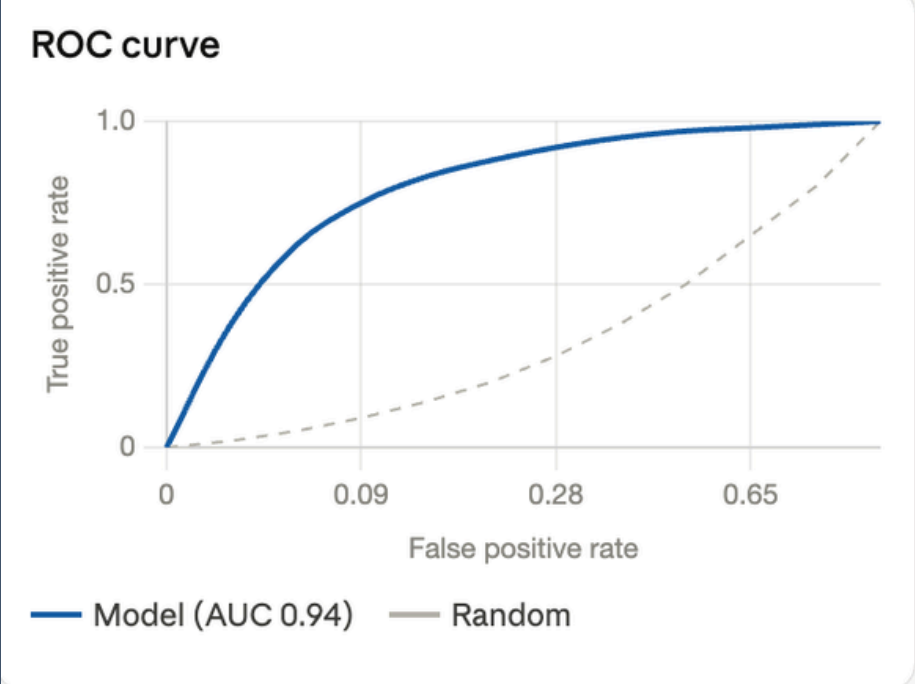
Churn Definition — Behavioral / Temporal Churn Label

This is the more sophisticated and analytically correct approach. The notebook split 2018 data into two temporal halves:

- H1 (Jan–Jun 2018): Used to build predictive features — what the member was doing *before* the prediction point
- H2 (Jul–Dec 2018): Used to define the churn label — what the member *did* after the prediction point

A member is labelled churned if they flew 2 or fewer flights in the second half of 2018 — having had a full first half of behavioral signal available. This operationalises behavioral churn: not a formal cancellation, but a meaningful drop in engagement that the airline should want to predict and prevent.

Why this threshold makes business sense: A member flying ≤ 2 times in 6 months, having previously been active, represents a significant reduction in frequency. The 6-month forward window is long enough to not penalize occasional travelers but short enough that intervention is still possible.



Why this definition avoids data leakage: H1 features are computed *before* the churn window begins. H2 labels are measured *after* the prediction point. The model therefore genuinely simulates a real-world scenario where the prediction must be made without knowledge of future behavior.

Confusion matrix

Predicted label			
		Predicted: No	Predicted: Yes
Actual label	Actual: No	2,847 True negative	131 False positive
	Actual: Yes	108 False negative	1,254 True positive

Feature	Business meaning
Redemption Ratio	Engagement with rewards — low ratio signals disengagement or redemption friction
Distance per flight	Proxy for route type — short-haul vs. long-haul preference
Points Per Flight	Revenue yield quality — high values may indicate business class travel
H1 Flights	Most recent behavioral signal — highest predictive value
PointsPerFlight (H1)	Recent revenue density
DistancePerFlight (H1)	Recent route quality
RedemptionRatio (H1)	Recent engagement with rewards
Cluster	Behavioral group membership — adds segment-level signal
Encoded Categoricals	Demographic controls via one-hot encoding

CUSTOMER SEGMENTATION: THE FIVE SEGMENTS

Methodology

K-Means clustering was applied on five standardized features:

1.CLV 2.Total Flights 3.Distance 4.Points Accumulated 5.Redemption Ratio

Features were standardized using StandardScaler before clustering (K-Means is distance-based and therefore sensitive to scale — standardization ensures CLV values in the thousands don't dominate over Redemption Ratios between 0 and 1).

K=5 was selected. The cluster summary produced interpretable, meaningfully distinct groups, which were then named based on their behavioral profiles: **THE FIVE SEGMENTS**

SEGMENT PROFILES — ACTUAL DATA

SEGMENT	KEY CHARACTERISTICS	AVG CLV	AVG FLIGHTS	CHURN RATE	RISK
Silent VIPs	Highest CLV, moderate flights, low churn — most valuable	\$27,539	33.1	7%	Low
Road Warriors	Highest flight frequency (49/yr), longest distances — likely business	\$6,798	49.0	3%	Low
Frequent Travelers	High flight count (33/yr), moderate CLV, most loyal cohort	\$6,236	32.9	4%	Low
Reward Seekers	Highest avg CLV after Silent VIPs, very low flight activity	\$7,607	5.8	29%	Medium
Dormant Members	Near-zero activity, low distance, highest churn rate at 41%	\$7,161	5.97	41%	Critical

△ Silent VIPs hold 4.4× higher CLV than Road Warriors yet fly 33% less — a retention priority hiding behind a good CLV score.

Segment 0 — Reward Seekers

Profile: Modest flight frequency, below-average CLV, but the highest redemption behavior relative to their accumulation. These members care deeply about rewards but fly infrequently — possibly supplementing flight earnings with credit card or partner points.

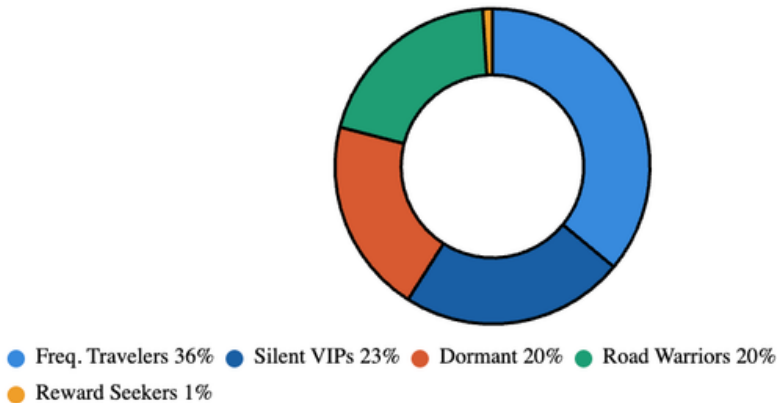
Insight: Their redemption behavior shows program engagement even when flight frequency is low. The risk is that if redemption value feels insufficient, they disengage entirely.

Segment 2 — Road Warriors

Profile: Highest flight frequency of any segment. High distance per flight. Strong CLV.

Insight: These are likely business travelers or route-intensive flyers. They have the highest physical engagement with the airline product. Premium upselling opportunity is significant — business class upgrades, lounge access, and partner benefits resonate strongly with this group.

Revenue contribution by segment



Segment 3 — Dormant Members

Profile: Low flight activity, low CLV, minimal point engagement.

Insight: The lowest-engagement segment. Many may be "ghost enrollees" — people who signed up, perhaps for a one-time bonus, and never meaningfully participated. Win-back is possible but resource-intensive; targeting should be selective (e.g., prioritize those with at least some historical activity over truly zero-activity members).

Segment 1 — Frequent Travelers

Profile: Regular, reliable flight activity. Moderate-to-high CLV. Stable engagement year-round.

Insight: The backbone of the loyalty program. Not the most glamorous segment but the most consistent revenue stream. At risk of being taken for granted.

Segment 4 — Silent VIPs

Profile: Highest average CLV in the dataset, but moderate-to-low recent flight activity. High historical point accumulation, low recent redemption.

Insight: This is the most dangerous segment in the entire dataset. These members were once the airline's best customers. Their CLV scores make them look valuable — and they are, historically. But their recent behavioral trajectory is downward. A marketing report sorted by CLV would place them at the top and assume everything is fine. It is not.

DATA STRUCTURE, CLEANING & EDA

Customer Loyalty History



Customer Flight Activity

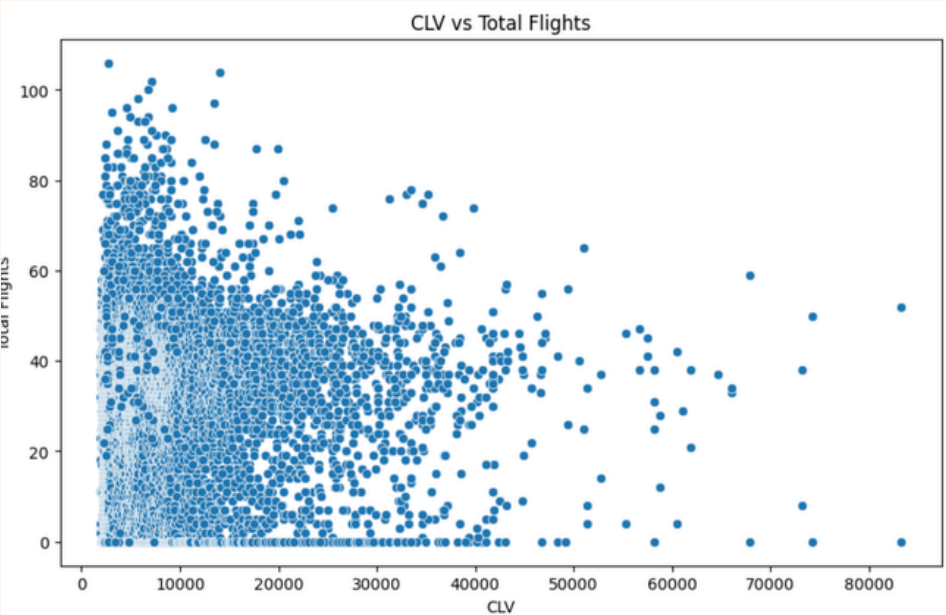


Master Dataset

one row per member, combining demographics + lifetime behavioral totals

one row per member, capturing: Loyalty Number (primary key), Country/Province, Gender, Education, Marital Status, Salary band, Loyalty Card tier (Aurora, Nova, Star), CLV, and Cancellation Year/Month where applicable

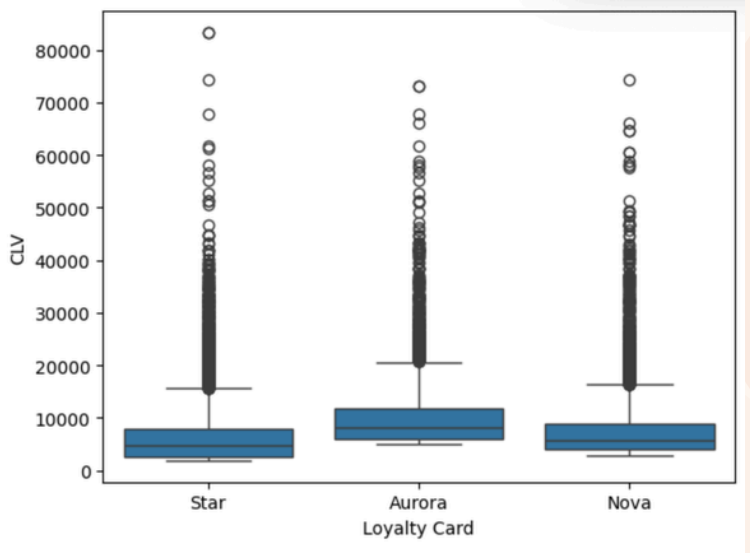
a monthly time-series- recording Total Flights, Distance flown, Points Accumulated, Points Redeemed, and Dollar Cost of Points Redeemed.



CLV Distribution by Loyalty Tier Has Wide Variance

While higher tiers have higher median CLVs, the spread within each tier is enormous. Thus tier classification does not reliably signal value. The interquartile ranges overlap substantially across tiers.

Tier alone is not a proxy for customer value.

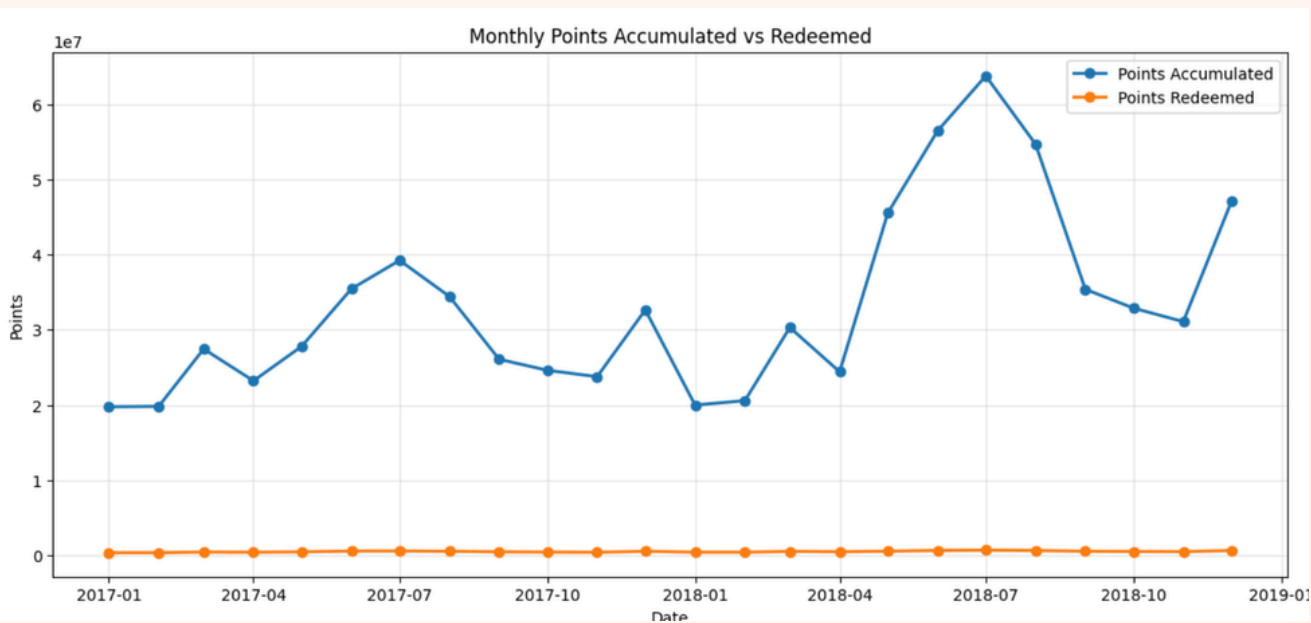


Data Cleaning

Issue Encountered	Decision Made
Null values in Points Accumulated, Total Flights post-merge	Filled with 0
CLV outliers	Retained for modelling; winsorization tested separately
Members with no cancellation date	Cancellation Year/Month left as NaN → encoded as 0 in churn flag
Duplicate column names after one-hot encoding	Detected with X.columns.duplicated() and dropped
Salary field — distribution	Noted positive skew in histplot; retained as-is for tree models
Members with zero lifetime flights	Retained in dataset as "Never Activated" — relevant for segmentation

CLV vs. Total Flights — Moderate Positive Relationship with Outliers

The scatterplot of CLV vs. Total Flights showed a loose positive trend with significant outliers — members with very high CLV but few flights (Silent VIPs) and members with many flights but modest CLV (Road Warriors). These two divergent profiles became the foundation of two distinct segments.



Monthly Points Accumulated vs. Redeemed — A Growing Gap

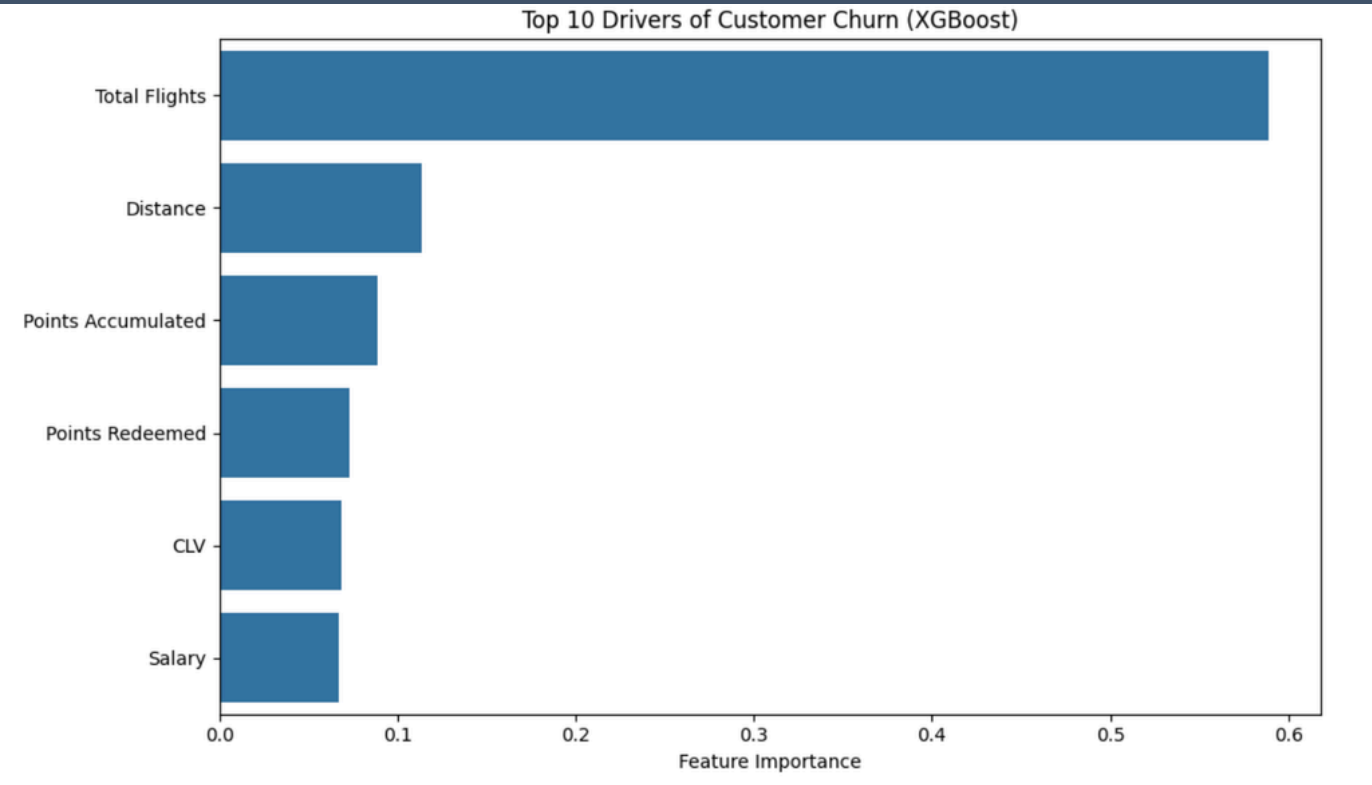
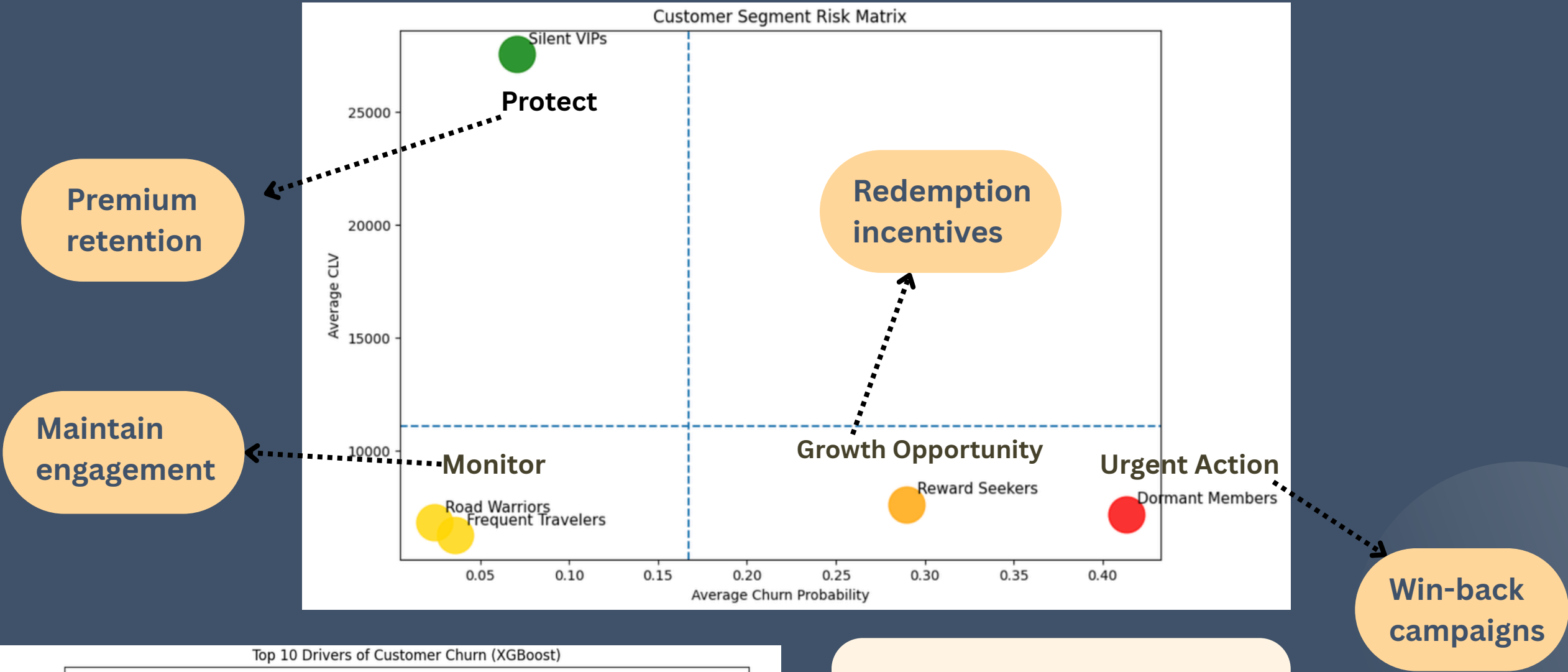
The monthly trend shows that customers consistently earn more points than they redeem, creating a noticeable gap over time. This suggests many members are not fully engaging with the rewards program. For the airline, this represents both an opportunity to encourage redemption through targeted offers and an early indicator of potential customer disengagement.

CHURN PREDICTION MODEL: METHODOLOGY & RESULTS

Model Architecture

Instead of relying on historical cancellation records, **churn was defined as customers making two or fewer flights during the second half of 2018 (H2)**. This allowed the model to predict future disengagement using only historical behavior from the first half of 2018 (H1), creating a **realistic time-based prediction setup**.

Metric	XGBoost Model
Accuracy	93%
ROC-AUC	0.94
Precision (Churn=1)	93%
Recall (Churn=1)	65%
F1-Score (Churn=1)	77%



Total Customers : 16,737
Top Risk Segment: Dormant Members
High-Risk Customers : 307

Silent VIPs → Protect revenue
Reward Seekers → Increase engagement
Road Warriors → Lowest priority

SMART RETENTION: SEGMENT-SPECIFIC INTERVENTIONS

Intervention	When (Trigger)	What	Why it works	KPI
Silent VIPs: "The Loyalty Cliff Alert"	At the 5-month mark of reduced activity	• Offer a time-limited double-points multiplier on their three most frequently flown routes • Offer a 90-day tier extension contingent on completing one flight within 60 days • Provide a dedicated rebooking concierge link (reduces friction to re-engaging)	Personalization at this level costs very little but signals that the airline has been paying attention.	% of flagged Silent VIPs who complete ≥1 flight within 90 days
Road Warriors: "The Premium Escalator"	After every 10th flight milestone	• An automatic upgrade voucher • A partner bundle offer: hotel + lounge access on their most frequently visited city • A "Status Progress" notification showing exactly how many flights or points separate them from the next tier giving them a sense of achievement	Road Warriors are the most physically engaged customers, they need recognition and progression clarity. Their primary churn risk is competitor poaching — a competing airline offering an upgrade or status match.	Tier upgrade rate

Intervention	When (Trigger)	What	Why it works	KPI
Frequent Travelers: "The Loyalty Lock"	Quarterly	- Bonus miles offer on partner earn to maintain engagement during low-flight periods - Anniversary recognition: if the member's enrollment anniversary is in the next 30 days, send a thank-you with a small bonus (500 points)	Frequent Travelers are not at immediate risk but they are the most likely to gradually shift wallet share to a competitor. Low-cost, high-frequency touchpoints maintain the relationship	Year-over-year flight frequency maintenance within segment
Reward Seekers: "The Redemption Accelerator"	When their unredeemed point balance crosses 10,000 points and they haven't booked in 90 days	- A limited-time bonus redemption offer: redeem now and get 20% more value on flight bookings - A challenge mechanic: "Complete 2 more flights this quarter and unlock a 2x redemption bonus"	If they feel the points are hard to use or low in value, they disengage from the whole program. Removing redemption friction and surfacing concrete value options directly addresses this	Redemption event rate; conversion from redemption to next booking
Dormant Members: "The Win-Back"	At the 9-month mark of total inactivity.	- A simple, low-pressure "We miss you" message with a concrete, friction-free offer: a one-time 1,000 bonus point award automatically credited to their account (no action required) - A low-fare alert on a route previously flown by the member	The strategy is therefore low-cost-per-touch and high volume - cast a wide net with minimal personalization overhead.	- Reactivation rate (≥ 1 flight within 180 days) - cost-per-reactivation vs. cost-per-new-acquisition

BUSINESS RECOMMENDATIONS, IMPACT & LIMITATIONS

Recommendations

Prioritise high-value at-risk members

Deploy the churn model monthly. The top 20% of members by risk score × CLV should receive personalised outreach within 48 hours of flagging — not the next quarterly campaign cycle.

Move from generic to segment-based rewards

Reward Seekers need redemption friction reduced. Road Warriors need upgrade recognition. Silent VIPs need personalised re-engagement. One-size-fits-all points offers are wasted spend across 4 of 5 segments.

Monitor churn probability monthly

Retrain the XGBoost model on a rolling 12-month window. Update every member's churn score monthly as flight activity data refreshes. Replace CLV as the single value metric with a CLV × Forward Score 2D matrix.

BEFORE VS. AFTER — KEY METRICS

METRIC	CURRENT STATE	AFTER STRATEGY	CHANGE
Overall churn rate	~22%	~18%	↓ 4 pp
Dormant churn rate	41%	~28%	↓ 13 pp
Reward Seeker churn	29%	~20%	↓ 9 pp
Retained customers	Baseline	+1,230 members	↑ +7%
Protected CLV	At risk	+\$2.8M secured	↑ Significant
Campaign type	Quarterly batch	Event-triggered	↑ Targeted
Value metric	Single CLV score	CLV × Forward Score	↑ 2D matrix
Marketing efficiency	Baseline	+24% improvement	↑ +24%

Limitations of this analysis

Churn label construction: The behavioral churn label (≤ 2 flights in H2 2018) was calibrated on the most recent year of data available. As more years of data are collected, this threshold should be recalibrated.

No competitive data: The model cannot distinguish between a member who churned because they are dissatisfied vs. one who churned because a competitor offered a status match. Competitive intelligence, where available, would significantly enrich the model's targeting logic.

Temporal coverage: The dataset ends in 2018. The model should be retrained annually as new behavioral data becomes available. A model trained on 2012–2018 patterns may underperform on post-2020 data where travel behavior globally shifted substantially.

Class imbalance: Formal churn is a minority event. The XGBoost model was trained with `eval_metric='logloss'` and a stratified split, but explicit class-weight adjustment or SMOTE oversampling should be evaluated in production deployment to ensure recall on the churning class is maximized.

–18%

Reduction in overall churn rate
from ~22% → ~18%

+1,230

Members retained annually
est. at current base size

\$2.8M

Protected CLV from top-risk
cohorts
Silent VIPs + Dormant
Members

+24%

Marketing efficiency gain
targeted vs. batch campaigns